

INTEGRATION

Standard results

- (1) $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ if $n \neq -1$
- (2) $\int 1 dx = x + C$
- (3) $\int \frac{1}{x} dx = \log x + C$ (or $\ln x + C$)
- (4) $\int \sqrt{x} dx = \frac{2}{3} x^{3/2} + C$
- (5) $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
- (6) $\int e^x dx = e^x + C$
- (7) $\int a^x dx = \frac{a^x}{\log a} + C$ where a is any constant
- (8) $\int \sin x dx = -\cos x + C$
- (9) $\int \cos x dx = \sin x + C$
- (10) $\int \sec^2 x dx = \tan x + C$
- (11) $\int \operatorname{cosec}^2 x dx = -\cot x + C$
- (12) $\int \sec x \cdot \tan x dx = \sec x + C$
- (13) $\int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + C$
- (14) $\int \tan x dx = \log(\sec x) + C$ (or $-\log(\cos x) + C$)
- (15) $\int \cot x dx = \log(\sin x) + C$
- (16) $\int \sec x dx = \log(\sec x + \tan x) + C$
- (17) $\int \operatorname{cosec} x dx = \log(\operatorname{cosec} x + \cot x) + C$
or
 $\log \tan \frac{x}{2} + C$

$$(18) \int (ax+b)^n dx = \frac{1}{a} \cdot \frac{(ax+b)^{n+1}}{n+1} + c \quad \text{if } n \neq -1$$

$$(19) \int \frac{1}{ax+b} dx = \frac{1}{a} \cdot \log(ax+b) + c$$

$$(20) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c$$

$$(21) \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + c$$

$$(22) \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + c$$

$$(23) \int (f(x))^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c ; n \neq -1$$

$$(24) \int \frac{f'(x)}{f(x)} dx = \log f(x) + c$$

$$(25) \int \phi(f(x)) f'(x) dx = \int \phi(t) dt \quad \text{where } t = f(x).$$

Integrate the following functions

(i) $\sqrt{2x+3}$ (ii) $\sin^2 x$ (iii) $\cos^3 x$

(iv) $\sin 2x \cos 4x$ (v) $\sin 5x \cdot \sin 2x$

(vi) $(x^2 + a^2)^3 x$ (vii) $\sin^7 x \cos x$

(viii) $\frac{\sin(\log x)}{x}$ (ix) $\frac{x^3}{(1+x^2)^5}$ (x) $\frac{1}{\sqrt{3x-2}}$

Answers

$$\begin{aligned} \text{(i)} \int (2x+3)^{1/2} dx &= \frac{1}{2} \cdot \frac{(2x+3)^{3/2}}{3/2} + c \\ &= \frac{1}{3} (2x+3)^{3/2} + c \end{aligned}$$

$$\begin{aligned} \text{(ii)} \text{ we know that } \sin^2 x &= \frac{1 - \cos 2x}{2} \\ \therefore \int \sin^2 x dx &= \int \frac{1 - \cos 2x}{2} dx = \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + c \end{aligned}$$

(iii) we know that

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\therefore 4 \cos^3 x = \cos 3x + 3 \cos x$$

$$\text{ie } \cos^3 x = \frac{1}{4} (\cos 3x + 3 \cos x)$$

$$\begin{aligned} \int \cos^3 x \, dx &= \frac{1}{4} \int (\cos 3x + 3 \cos x) \, dx \\ &= \frac{1}{4} \left[\frac{\sin 3x}{3} + 3 \sin x \right] + C \end{aligned}$$

(iv) we know that

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$\begin{aligned} \therefore \sin 2x \cos 4x &= \frac{1}{2} (2 \sin 2x \cos 4x) \\ &= \frac{1}{2} [\sin 6x + \sin(-2x)] \\ &= \frac{1}{2} [\sin 6x - \sin 2x] \end{aligned}$$

$$\begin{aligned} \therefore \int \sin 2x \cos 4x \, dx &= \frac{1}{2} \int (\sin 6x - \sin 2x) \, dx \\ &= \frac{1}{2} \left[-\frac{\cos 6x}{6} + \frac{\cos 2x}{2} \right] + C \end{aligned}$$

(v) we know that

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\begin{aligned} \therefore \sin 5x \sin 2x &= \frac{1}{2} (2 \sin 5x \sin 2x) \\ &= \frac{1}{2} (\cos(3x) - \cos 7x) \end{aligned}$$

$$\begin{aligned} \int \sin 5x \sin 2x \, dx &= \frac{1}{2} \int (\cos 3x - \cos 7x) \, dx \\ &= \frac{1}{2} \left[\frac{\sin 3x}{3} - \frac{\sin 7x}{7} \right] + C \end{aligned}$$

$$(vi) \int (x^2 + a^2)^3 x \, dx = \int u^3 \frac{du}{2}$$

$$= \frac{1}{2} \frac{u^4}{4} + C$$

$$= \frac{(x^2 + a^2)^4}{8} + C$$

$$\begin{array}{l} \text{Put } u = x^2 + a^2 \\ du = 2x \, dx \\ \frac{du}{2} = x \, dx \end{array}$$

$$\begin{aligned}
 \text{(vii)} \quad \int \sin^7 x \cos x \, dx &= \int u^6 \, du && \begin{pmatrix} u = \sin x \\ du = \cos x \, dx \end{pmatrix} \\
 &= \frac{u^7}{7} + C \\
 &= \frac{\sin^7 x}{7} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii)} \quad \int \frac{\sin(\log x)}{x} \, dx &= \int \sin(\log x) \cdot \frac{1}{x} \, dx && \begin{pmatrix} \text{Put } u = \log x \\ du = \frac{1}{x} \, dx \end{pmatrix} \\
 &= \int \sin u \cdot du \\
 &= -\cos u + C \\
 &= -\cos(\log x) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ix)} \quad \int \frac{x^3}{(1+x^2)^5} \, dx &= \int \frac{x^2}{(1+x^2)^5} x \, dx \\
 &= \int \frac{u-1}{u^5} \frac{du}{2} \\
 &= \frac{1}{2} \int \left(\frac{1}{u^4} - \frac{1}{u^5} \right) du \\
 &= \frac{1}{2} \int (u^{-4} - u^{-5}) \, du \\
 &= \frac{1}{2} \left[\frac{u^{-3}}{-3} - \frac{u^{-4}}{-4} \right] + C \\
 &= \frac{1}{2} \left[-\frac{1}{3u^3} + \frac{1}{4u^4} \right] + C \\
 &= \frac{1}{2u^4} \left[-\frac{u}{3} + \frac{1}{4} \right] + C \\
 &= \frac{1}{2(1+x^2)^4} \left[-\frac{(1+x^2)}{3} + \frac{1}{4} \right] + C
 \end{aligned}$$

Put $u = 1+x^2$
 $du = 2x \, dx$
 $x \, dx = \frac{du}{2}$
 Also $x^2 = u-1$

$$\begin{aligned}
 \text{(x)} \quad \int \frac{1}{\sqrt{3x-2}} \, dx &= \int (3x-2)^{-1/2} \, dx \\
 &= \frac{1}{3} \frac{(3x-2)^{1/2}}{1/2} + C \\
 &= \frac{2}{3} \sqrt{3x-2} + C
 \end{aligned}$$

Some more standard results

- (1) $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- (2) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$
- (3) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$
- (4) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
- (5) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log(x + \sqrt{x^2+a^2}) + C$
- (6) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log(x + \sqrt{x^2-a^2}) + C$
- (7) $\int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$
- (8) $\int \sqrt{x^2+a^2} dx = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \log(x + \sqrt{x^2+a^2}) + C$
- (9) $\int \sqrt{x^2-a^2} dx = \frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \log(x + \sqrt{x^2-a^2}) + C$

Integrate the following

- (i) $\frac{1}{x^2+2x+2}$ (ii) $\frac{1}{25x^2-16}$ (iii) $\frac{1}{\sqrt{81+64x^2}}$
- (iv) $\frac{1}{\sqrt{8-2x-x^2}}$ (v) $\sqrt{4-x^2}$

Answers

$$\begin{aligned}
 \text{(i)} \quad \int \frac{dx}{x^2+2x+2} &= \int \frac{dx}{(x+1)^2+1^2} \\
 &= \frac{1}{1} \tan^{-1}\left(\frac{x+1}{1}\right) + C \\
 &= \tan^{-1}(x+1) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int \frac{dx}{25x^2-16} &= \int \frac{dx}{25\left(x^2-\frac{16}{25}\right)} \\
 &= \frac{1}{25} \int \frac{dx}{x^2-\left(\frac{4}{5}\right)^2}
 \end{aligned}$$

$$= \frac{1}{25} \cdot \frac{1}{2 \times \frac{4}{5}} \log \left| \frac{x - \frac{4}{5}}{x + \frac{4}{5}} \right| + C$$

$$= \frac{1}{40} \log \left| \frac{5x-4}{5x+4} \right| + C$$

$$\begin{aligned} \text{(iii)} \int \frac{1}{\sqrt{81+64x^2}} dx &= \int \frac{dx}{\sqrt{64\left(\frac{81}{64}+x^2\right)}} \\ &= \frac{1}{8} \int \frac{dx}{\sqrt{\left(\frac{9}{8}\right)^2+x^2}} \\ &= \frac{1}{8} \log \left[x + \sqrt{\left(\frac{9}{8}\right)^2+x^2} \right] + C \\ &= \frac{1}{8} \log \left[x + \sqrt{\frac{81}{64}+x^2} \right] + C \end{aligned}$$

$$\begin{aligned} \text{(iv)} \int \frac{dx}{\sqrt{8-2x-x^2}} &= \int \frac{dx}{\sqrt{8-(x^2+2x)}} \\ &= \int \frac{dx}{\sqrt{9-(x^2+2x+1)}} \\ &= \int \frac{dx}{\sqrt{3^2-(x+1)^2}} \\ &= \sin^{-1} \left(\frac{x+1}{3} \right) + C \end{aligned}$$

$$\begin{aligned} \text{(v)} \int \sqrt{4-x^2} dx &= \int \sqrt{2^2-x^2} dx \\ &= \frac{x}{2} \sqrt{2^2-x^2} + \frac{2^2}{2} \sin^{-1} \left(\frac{x}{2} \right) + C \\ &= \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) + C \end{aligned}$$

Assignment problems

Integrate

(i) $\sqrt{3-2x}$

(iii) $\sin^3 x$

(v) $x^2 \cos(x^3+1)$

(vii) $\frac{1}{\sqrt{x^2+3x+1}}$

(ix) $\sqrt{9-36x^2}$

(ii) $\cos^2 x$

(iv) $\cos 5x \cdot \cos x$

(vi) $e^{\tan x} \sec^2 x$

(viii) $\frac{1}{x^2+4x+13}$

Integration by parts

This is a method to integrate product of two functions.

$$\int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

$$\text{or } \int \text{I} \cdot \text{II} = \text{I} \int \text{II} - \int [d(\text{I}) \int \text{II}]$$

Integrate the following

(i) $x \cos x$ (ii) $x^2 \sin x$ (iii) $x \log x$

(iv) $\tan^{-1} x$ (v) $x^2 e^{3x}$

$$\begin{aligned} \text{(i)} \quad \int \underset{\text{I}}{x} \underset{\text{II}}{\cos x} dx &= \text{I} \int \text{II} - \int (d(\text{I}) \int \text{II}) \\ &= x \sin x - \int 1 \cdot \sin x dx \\ &= x \sin x - (-\cos x) + C \\ &= x \sin x + \cos x + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int \underset{\text{I}}{x^2} \underset{\text{II}}{\sin x} dx &= x^2 (-\cos x) - \int 2x (-\cos x) dx \\ &= -x^2 \cos x + 2 \int \underset{\text{I}}{x} \underset{\text{II}}{\cos x} dx \\ &= -x^2 \cos x + 2 \left[x \sin x - \int 1 \cdot \sin x dx \right] \\ &= -x^2 \cos x + 2 \{ x \sin x - (-\cos x) \} + C \\ &= (2 - x^2) \cos x + 2x \sin x + C \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \int \underset{\text{I}}{x} \underset{\text{II}}{\log x} dx &= \log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \\ &= \frac{x^2}{2} \log x - \frac{1}{2} \int x dx \\ &= \frac{x^2}{2} \log x - \frac{1}{2} \left(\frac{x^2}{2} \right) + C \\ &= \frac{x^2}{2} \left(\log x - \frac{1}{2} \right) + C \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \int \underset{\text{I}}{\tan^{-1} x} \underset{\text{II}}{1} dx &= \int \tan^{-1} x \cdot \frac{1}{x} dx \\ &= \tan^{-1} x \cdot x - \int \frac{1}{1+x^2} x dx \end{aligned}$$

$$= x + \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= x + \tan^{-1} x - \frac{1}{2} \log(1+x^2) + C$$

$$\begin{aligned} \text{(v)} \quad \int \underbrace{x^2}_{\text{I}} \underbrace{e^{3x}}_{\text{II}} dx &= \frac{x^2 e^{3x}}{3} - \int 2x \frac{e^{3x}}{3} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int \underbrace{x}_{\text{I}} \underbrace{e^{3x}}_{\text{II}} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[x \frac{e^{3x}}{3} - \int 1 \cdot \frac{e^{3x}}{3} dx \right] \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[\frac{x e^{3x}}{3} - \frac{1}{3} \frac{e^{3x}}{3} \right] + C \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} \\ &= \frac{e^{3x}}{27} [9x^2 - 6x + 2] \end{aligned}$$

short cut method

$$\int u v dx = u v_1 - u' v_2 + u'' v_3 - \dots$$

where $u' = \frac{du}{dx}$, $u'' = \frac{d^2 u}{dx^2}$; $v_1 = \int v dx$, $v_2 = \int v_1 dx$, $\dots$

$$\begin{aligned} \text{(i)} \quad \int x \cos x dx &= x \sin x - 1 \cdot (-\cos x) + C \quad \left(\begin{array}{l} u = x \\ v = \cos x \end{array} \right) \\ &= x \sin x + \cos x + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int \underbrace{x^2}_u \underbrace{\sin x}_v dx &= x^2 (-\cos x) - 2x (-\sin x) + 2 \cdot (\cos x) + C \\ &= (2 - x^2) \cos x + 2x \sin x + C \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \int \underbrace{x^2}_u \underbrace{e^{3x}}_v dx &= x^2 \cdot \frac{e^{3x}}{3} - 2x \frac{e^{3x}}{9} + 2 \cdot \frac{e^{3x}}{27} + C \\ &= \frac{e^{3x}}{27} [9x^2 - 6x + 2] + C \end{aligned}$$

Integrate

(i) $x \sin x$

(ii) $x^2 \cos x$

(iii) $x^3 e^{2x}$

(iv) $\log x$

(v) $\sin^{-1} x$

Two more results

$$(1) \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$(2) \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

Evaluate (i) $\int e^{2x} \sin 3x \, dx$ (ii) $\int e^{3x} \cos x \, dx$

(i) $a=2, b=3$

$$\begin{aligned} \therefore \int e^{2x} \sin 3x \, dx &= \frac{e^{2x}}{2^2 + 3^2} (2 \sin 3x - 3 \cos 3x) + C \\ &= \frac{e^{2x}}{13} (2 \sin 3x - 3 \cos 3x) + C \end{aligned}$$

(ii) $a=3, b=1$

$$\begin{aligned} \therefore \int e^{3x} \cos x \, dx &= \frac{e^{3x}}{3^2 + 1^2} (3 \cos x + 1 \cdot \sin x) + C \\ &= \frac{e^{3x}}{10} (3 \cos x + \sin x) + C \end{aligned}$$

Integrate (i) $e^{3x} \sin x$ (ii) $e^x \cos 2x$